FragrAI: An AI-Powered Fragrance Recommendation Engine

Using TF-IDF, cosine similarity, and GPT-4o to match users with fragrances from a dataset of 84,000+

75%
VALIDATION ACCURACY

INTRODUCTION

Problem

Shopping for a fragrance is overwhelming. Customers typically buy blind — relying on word-of-mouth or in-store staff — because there is no way to truly know how a scent will perform before purchase. Existing databases like **Fragrantica** catalog tens of thousands of perfumes but have cluttered interfaces and no intelligent recommendations.

Significance

Fragrance influences confidence, mood, and social interaction by acting on the limbic system, triggering dopamine and serotonin release (Gupta, 2025; Sowndhararajan & Kim, 2016). A well-matched fragrance can boost self-assurance — but only if it's the *right* match.

Objective

Build a full-stack AI recommendation system combining an ML backend with a React website, delivering fragrances tailored to each user's preferences and use case.

BACKGROUND

- **TF-IDF** (Term Frequency–Inverse Document Frequency) converts text into numerical vectors for comparison.
- **Cosine similarity** measures alignment between two vectors — closer to 1.0 means a stronger match.
- **OpenAI GPT-4o** interprets natural-language descriptions and extracts meaningful scent preferences.
- Prior work: Santana et al. (2021) used deep neural networks to formulate fragrances via Particle Swarm Optimization; Lötsch et al. (2018) showed ML can predict odor perception from molecular structure.

$$similarity(A, B) = (A \cdot B) / \|A\| \|B\|$$

HYPOTHESIS

A machine-learning model trained on a large fragrance dataset can accurately recommend personalized fragrances by combining **TF-IDF vectorization**, **cosine similarity**, and **natural language processing** of user descriptions.

SYSTEM ARCHITECTURE

Backend

Python · FastAPI · Scikit-learn · Pandas · OpenAI API

Frontend

React · TypeScript · Vite · Tailwind CSS

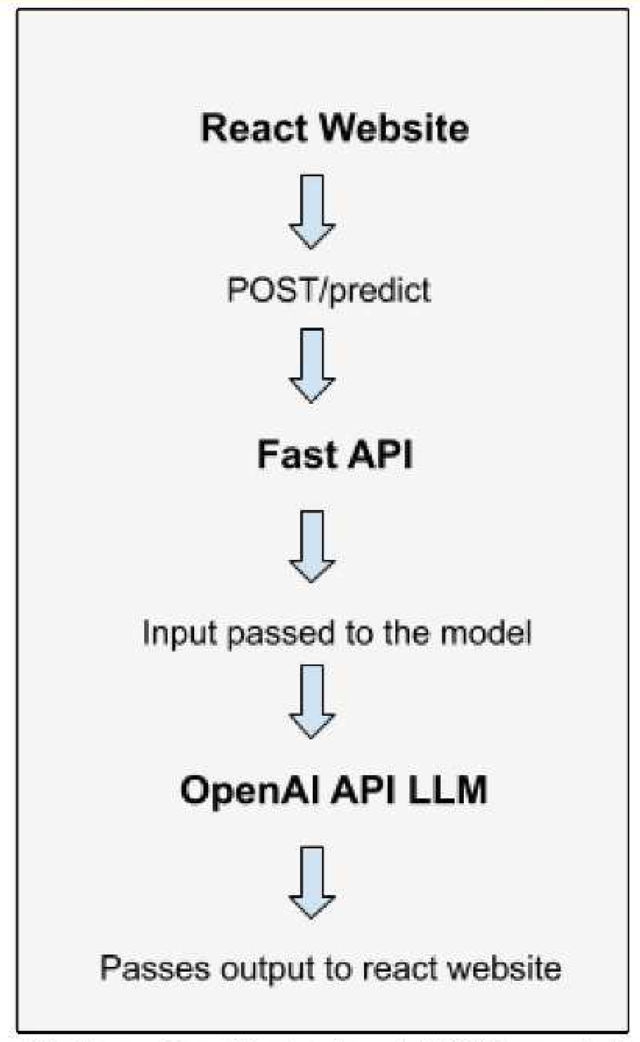

Figure 1. *System architecture — React front-end sends POST requests through FastAPI to the ML model and OpenAI API.*

METHODS — DATASET

84,000+ fragrances sourced from a public Kaggle dataset scraped from Fragrantica, including notes (top/middle/base), designer, gender, and seasonal classification. The data was cleaned and classified by season using a TF-IDF + SVM pipeline.

	title	designer	Gender	predicted_season_tfidf_svm	top_notes	middle_notes	base_notes
1709	Interlude Man Amouage for men	Amouage Perfumes and Colognes	Male	Winter	[Oregano, Pepper, Bergamot]	[Incense, Opoponax, Amber, Labdanum]	[Agarwood Oud, Leather, Sandalwood, Patchouli]
7665	Nothing More for Men Gosh for men	Gosh Perfumes and Colognes	Male	Winter	[Marine, Bergamot, Lime]	[Lemon, White Pepper, Cardamom, Jasmine]	[Sage, Leather, Sandalwood, Amber]
4800	Modanité Godet for men	Godet Perfumes and Colognes	Male	Winter	[Rose, Pepper, Patchouli, White Tobacco]	[Tobacco, Osmanthus, Tonka Bean, Iris]	[Vanilla, Leather, Sandalwood, Cedar]

Figure 2. *Sample of the cleaned dataset with predicted season and note structure.*

Validation

An **800-train / 200-test** split was performed on fragrances with existing user reviews, achieving **75% validation accuracy**.

RECOMMENDATION PIPELINE

1. User completes a **4-step questionnaire**: gender, age, season, use case, budget + a free-text description.
2. **GPT-4o** extracts scent preferences and key notes from the free-text input.
3. **TF-IDF** vectorizes fragrance notes (3,000 dim); seasons one-hot encoded (5 dim).
4. Feature matrix weighted **85% note similarity + 15% seasonal compatibility**.
5. **Cosine similarity** computed against all 84,000+ fragrances; brand boost applied to 1,083 Tier-1 designers.
6. Top 3 matches filtered by gender & threshold, returned with match scores and shopping links.

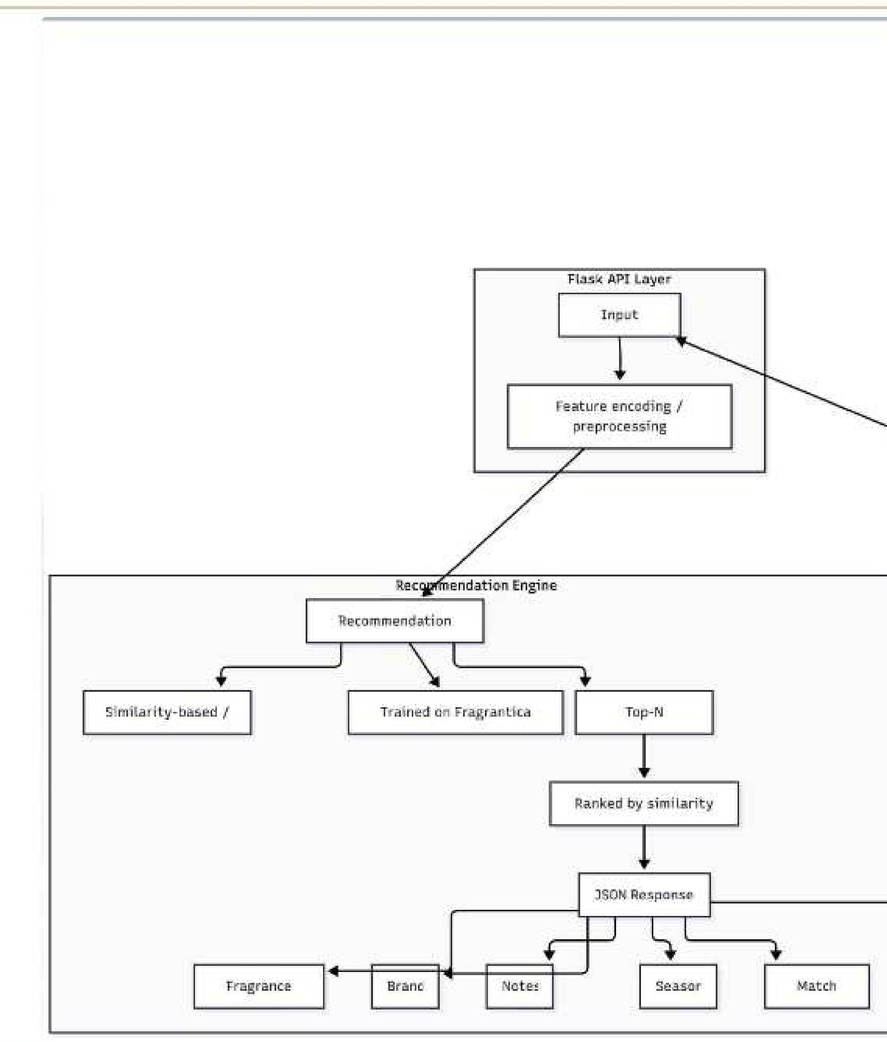

Figure 3. *Recommendation engine flowchart — similarity-based ranking returns a Top-N JSON response.*

USER INTERFACE FLOW

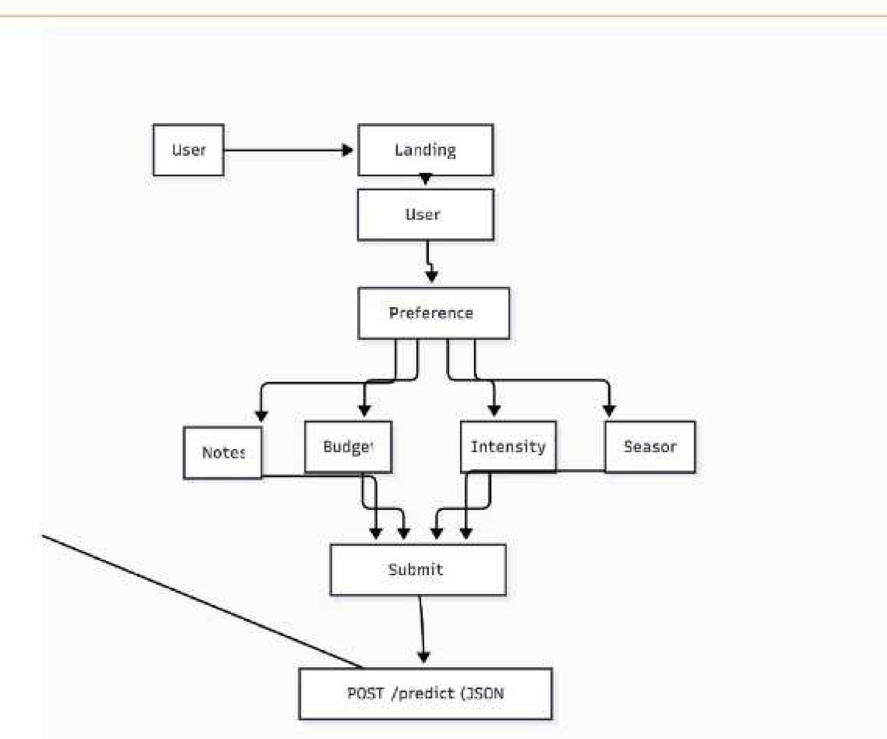

Figure 4. *User flow: landing › preference questionnaire (notes, budget, intensity, season) › submit → JSON recommendation.*

RESULTS & DISCUSSION

75% VALIDATION ACCURACY	**62–67%** LIVE MATCH RANGE
< 3s RESPONSE TIME	**84,144** FRAGRANCES INDEXED

FragrAI accepts user input through the web interface and returns three personalized recommendations with match scores, notes, season, and shopping links. The **800-train / 200-test** validation achieved **75% accuracy**.

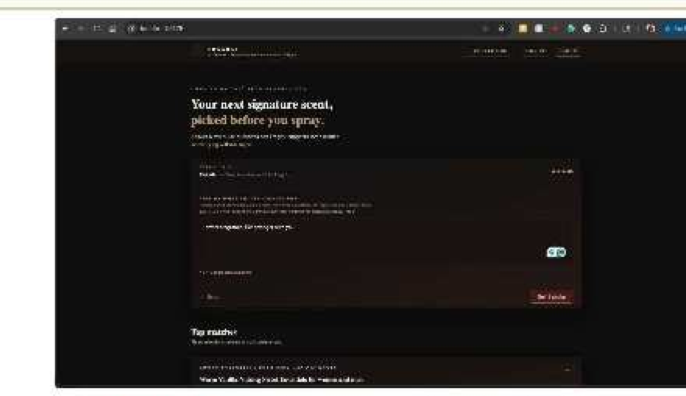

Figure 5. *Landing page + 4-step questionnaire.*

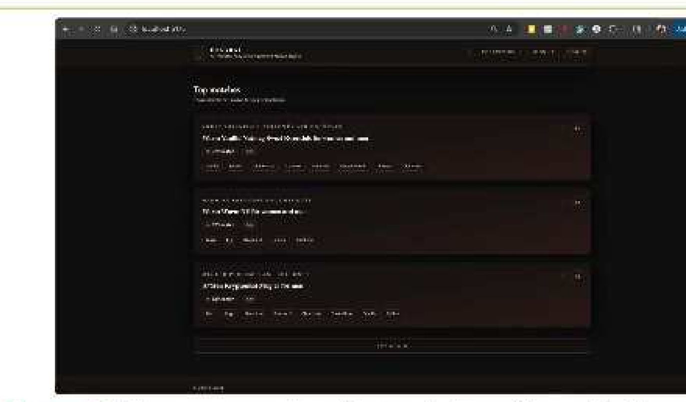

Figure 6. *Three personalized top matches with match % and notes.*

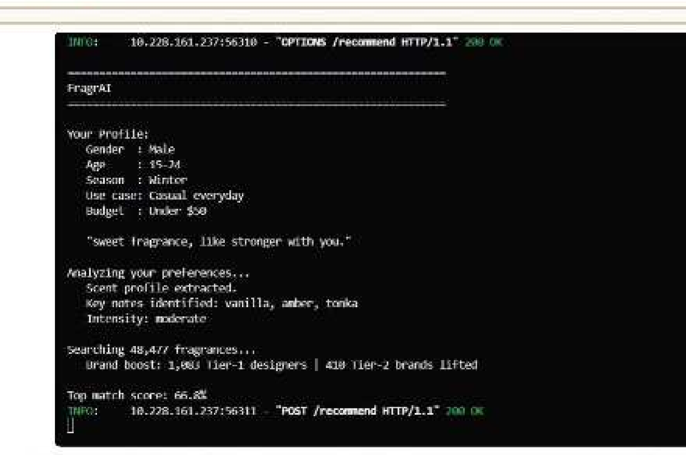

Figure 7. *Backend log — GPT-4o extracts notes and intensity.*

Discussion

A consistent **60–67% live match score** and **75% validation accuracy** represent meaningful personalization — well above random or keyword baselines. The hybrid **TF-IDF + GPT-4o** approach proved effective: classical vectorization handles structured note data, while the LLM bridges free-text descriptions and the fragrance feature space.

Future Work

- Train a **custom embedding model** on fragrance-specific language.
- Add a **user feedback loop** for continuous learning.
- Expand with **price, longevity, and sillage** metrics.
- Deploy on a **cloud-hosted backend**.
- Add **collaborative filtering**.

REFERENCES

Gupta, S. (2025). What is the Connection Between Fragrance, Mood, and Confidence? *ENVY*.

Lötsch, J., Kringel, D., & Hummel, T. (2018). Machine learning in human olfactory research. *Chemical Senses, 44*(1), 11–22.

Santana, V. V., et al. (2021). Optimal fragrances formulation using a deep learning neural network architecture. *Computers & Chemical Engineering, 150*, 107344.

Sowndhararajan, K., & Kim, S. (2016). Influence of Fragrances on Human Psychophysiological Activity. *Scientia Pharmaceutica, 84*(4), 724–751.

Fragrantica data (2024). Kaggle Dataset.